

Coherent Passive Optical Networks for 100G/ λ -and-Beyond Fiber Access: Recent Progress and Outlook

Junwen Zhang and Zhensheng Jia

ABSTRACT

Coherent optics is considered a promising candidate for realizing single-wavelength passive optical networks (PONs) at 100 G/ λ and beyond. It has been a game changer for enabling ultra-high-speed data transmission in long-haul and metro networks. Interest in coherent optics is at an all-time high for application to short-reach access networks. The use of a local oscillator in coherent beating with received signals results in superior receiver sensitivity and thereby an extended power budget that supports high-capacity, long-reach, and large-split-ratio access. It has also been demonstrated to be beneficial for flexible deployment and adaptive access by advanced modulation formats and powerful digital signal processing. In this article, we review use cases, challenges, and recent progress in high-speed coherent PONs for 100 G/ λ and beyond. We also provide technical insights into key enabling technologies in coherent PONs, including innovations for simplification and optimization of coherent optics and coherent upstream burst-mode detection. Finally, we discuss the outlook for coherent PONs toward realizing a rate-flexible and adaptive fiber access network. The architecture of a rate-flexible single-wavelength time-and-frequency-division multiplexing coherent PON based on digital subcarrier multiplexing technology is introduced. In addition, the adaptive fiber access network enabled by the adaptive modulation and coding scheme is also discussed.

INTRODUCTION

The passive optical network (PON) is a point-to-multipoint (P2MP) broadband fiber access technology that has successfully been demonstrated as a low-cost and effective solution for the “last mile” connection between service providers and customers over all passive optical distribution networks (ODNs) [1, 2]. PONs have been deployed worldwide since the 1990s to support fiber-to-X (FTTX) technology, where X can represent home (FTTH), office (FTTO), building (FTTB), mobile cell (FTTC), and so on. So far, residential broadband access services have been the main PON applications. Advances in the high-speed optical access network have always been propelled by novel business and application drivers [2]. To meet rising bandwidth demands, several generations of PON systems have been standardized through

the efforts of two major industry standardization organizations: the International Telecommunication Union Telecommunication Standardization Sector (ITU-T)/Full Service Access Network (FSAN) Group and the IEEE 802.3 Ethernet Working Group. Two major types of PON standards, namely, GPON and Ethernet PON (EPON), have been developed by ITU-T/FSAN and IEEE 802.3, respectively. Figure 1 illustrates the evolution in PON standards in terms of the maximum provided bandwidth from broadband PON (B-PON) to different generations of PON systems. The figure shows that over the last few decades, the PON access speed over a single wavelength has increased by almost 100 times from hundreds of megabits per seconds per wavelength to the current speed of 25/50 Gb/s per wavelength [3].

Driven by emerging services, such as 5G mobile Internet, cloud networking, and high-definition (HD) video streaming services, the optical access network industry is embracing a new wave of bandwidth acceleration. This change is further evidenced by recent progress by standards bodies in developing next-generation high-speed PON standards. To support the ever-growing bandwidth demand from these emerging services, the 25/50G next-generation Ethernet PON (NG-EPON) specification was approved by IEEE in 2020 based on 25 Gb/s per single wavelength, and the 25GS-PON multi-source agreement (MSA) is under testing. The ITU-T 50G PON standard (also referred to as G.9804), based on a 50G single wavelength, is now close to becoming a reality because consent was obtained for multiple parts of the series in 2021.

It is envisioned that 100 Gb/s per wavelength and beyond will be required in the future to meet continuously growing bandwidth demands. Intensity modulation and direct detection (IM/DD) technologies have been used in the physical layer of previous generations of PON systems ranging from B-PON to the current 50G PON. IM/DD, which employs electro-absorption modulated laser (EML) or directly modulated laser (DML) at the transmitter to modulate signals on the optical intensity and pin photodiodes (PIN) or avalanche photodiode (APD) at the receiver to directly detect the optical intensity, is an attractive approach due to its cost effectiveness and simple configuration. However, many questions and challenges must be resolved for high-speed

access networks to support 100G based on the IM/DD achieving the same power budgets as current systems [3]. One solution is the use of multiple wavelengths of IM/DD signals, namely, wavelength-division multiplex (WDM) or dense WDM (DWDM), to achieve 100G capacity. For example, 100G could be aggregated by wavelength bonding of four 25G channels.

Alternatively, a high-speed PON based on a single wavelength with a time-division multiplex (TDM) mechanism can reduce the number of required optical components and the related cost for colorless modules and therefore remains of considerable interest to both vendors and operators. In addition, it saves wavelength resources and potentially reduces the operation costs. For example, 100G PONs with wavelength multiplexing and IM/DD of four 25 Gb/s or two 50 Gb/s channels are viewed as too challenging because of a limited power budget and complex wavelength resource management. Finally, the single-wavelength line rate will always be the baseline rate for future WDM or TWDM-PON systems, as developed in IEEE and ITU-T. An example is the ITU-T 50G-PON system with three subtypes: 50G TDM PON, 50 Gb/s/ TWDM PON, and 50 Gb/s/ point-to-point WDM PON (PtP WDM PON). Therefore, how to achieve 100G and beyond in a single-wavelength PON has become an active research area for next-generation optical access networks in the post 50G-PON era and has stimulated considerable discussion in both industry and academia.

Will coherent optics be an answer for next-generation high-speed optical access networks? In this article, we attempt to answer this question through a comprehensive review of recent research on coherent PONs.

In the next section, we review the motivations for employing coherent optics for 100-Gb/s-and-beyond single-wavelength fiber access by considering three major application scenarios and the four most important advantages. We analyze the bandwidth requirements for these application scenarios and show how coherent optics can support 100G-and-beyond PONs with advantages beyond sensitivity improvement. Following that, we analyze two major challenges posed by coherent PONs, including the challenge of cost, complexity, and power consumption, and upstream burst-mode coherent detection. Then we provide technical insight into the two key enabling technologies for coherent PONs: the simplification and optimization of coherent optics, and burst-mode coherent detection. Following that, we discuss the outlook for coherent PONs, including future research directions for flexible multiplexing beyond TDM and rate-adaptive coherent access. Finally, we conclude the article.

Note that a coherent WDM PON or an ultra DWDM (UDWDM) PON with additional capability of frequency selection from coherent detection has been widely discussed and studied [4]. However, the objective of this article is not to debate whether future coherent PONs should be TDM or WDM, but rather to explore how technologies need to evolve to keep pace with the continued growth of speed for a single wavelength. Therefore, as discussed above, this article focuses on the evolution of single-wavelength 100G-and-beyond PON systems.

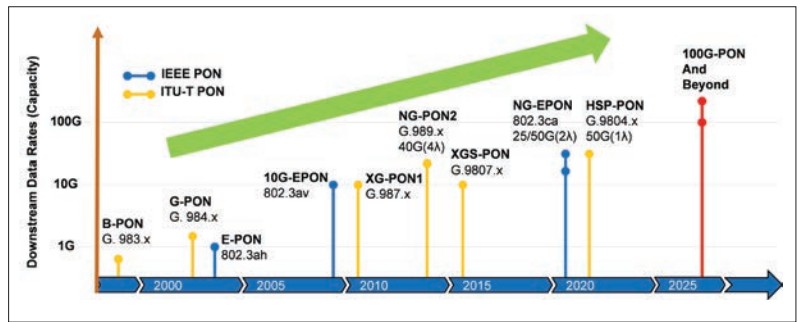


FIGURE 1. The evolution of the PON from B-PON to 1G/λ, 10G/λ, 25G/λ, 50G/λ and future 100G/λ and beyond.

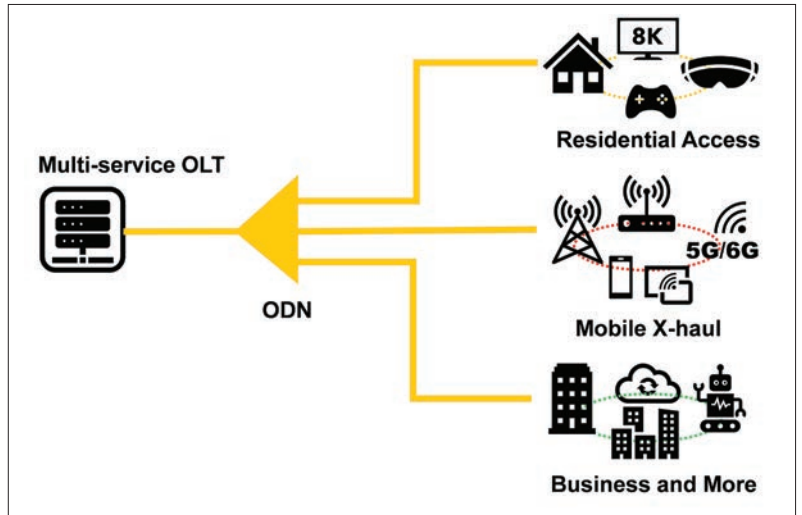


FIGURE 2. Use scenarios for future high-speed PON systems, showing three major types of application scenarios.

MOTIVATIONS

APPLICATION SCENARIOS

Residential broadband access service is the major application scenario for PONs, and has been widely discussed and studied; however, the use cases of PONs have currently been extended considerably beyond residential services [2]. Figure 2 shows use scenarios for future PONs, where multiple types of services could be provided by future high-speed PON systems, including residential services, mobile X-haul, and business services. In this section, we analyze the bandwidth requirements for these services and show how 100G-and-beyond PONs will be needed.

Residential Broadband Access: Residential broadband access rates are growing rapidly worldwide thanks to technologies such as high-quality video streaming services such as 8K/16K video, virtual reality/augmented reality (VR/AR), and camera-based smart home services. Over-the-top video service and cloud gaming continuously demand high-quality video streaming and therefore consume increasing bandwidth in the access network. A general estimate of the downlink bandwidth is 1.4 Gb/s for advanced VR in 3–5 years, increasing to 3–4 Gb/s for extreme or ultimate VR in 6–10 years. Considering a 1:64 split ratio, the overall downlink bandwidth requirement of 64 subscribers could be over 84 Gb/s considering a 50 percent actual installation rate and a 75 percent concurrent rate for 3.5 Gb/s

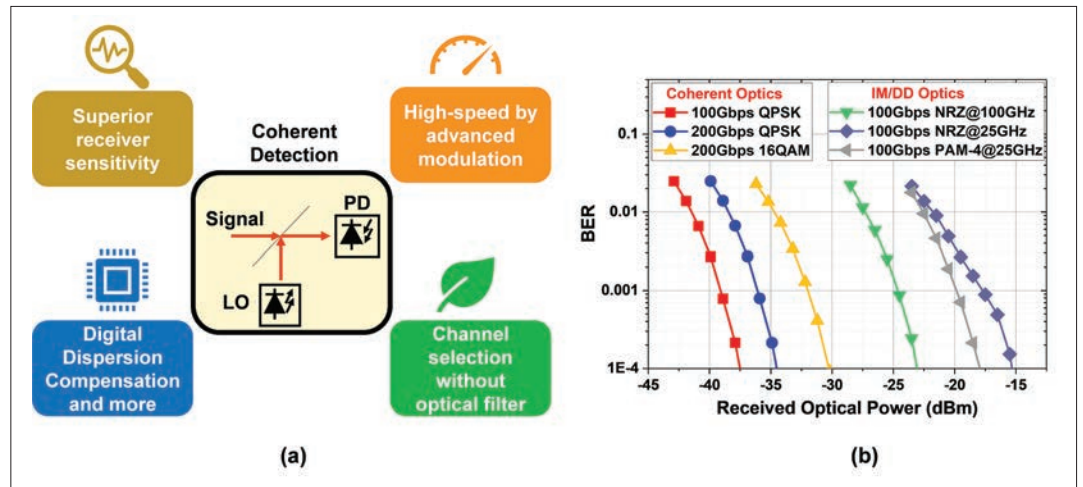


FIGURE 3. a) Four advantages of using coherent optics in an access network; b) comparison of the sensitivity of coherent optics and IM/DD optics for 100-Gb/s and 200-Gb/s line rates. Simulation setup in b): Coherent Optics using full-field, polarization-/phase-diversity, and homodyne coherent detection, the responsivities of PINs are all set at 0.7 A/W, thermal noise level is set at $10e^{-12}$ A/(Hz) $^{1/2}$; IM/DD Optics using EML with extinction ratio (ER) of 8dB, and APD with responsivity of 0.7 A/W, avalanche multiplication factor of 13, and ionization coefficient of 0.13, thermal noise level of $10e^{-12}$ A/(Hz) $^{1/2}$.

ultimate VR services (84 Gb/s = 3.5 Gb/s $\times$ 64 $\times$ 0.5 $\times$ 0.75). Therefore, a 100G PON in approximately 2030 could foreseeably support bandwidth requirements.

Mobile X-haul: Mobile x-haul has become one of the largest market drivers for new generations of PON systems [5]. Considering the massive market and rapid growth of radio access networks (RANs), the FSAN group began a project in 2016 to study the role of high-speed PONs in RAN evolutions in the released FSAN Standards Roadmap 2.0. 5G x-haul is a development of 4G fronthaul that separates data processing within a distributed unit (DU) from a signal-bearing radio unit (RU). Large-scale deployment of dense RU networks poses major infrastructure challenges while creating enormous opportunities for the use of optical access networks. As a P2MP topology for the efficient use of fiber resources and its wide deployment, PON stands out as a strong candidate as a cost-effective solution for 5G x-haul links. The calculated peak downlink bandwidth for the sub-6 GHz frequency ranges from 3.024 Gb/s (F1 interface) to 86.1 Gb/s (Fx interface), depending on the choice of specific function split options, assuming a 100 MHz radio frequency bandwidth, 256-quadrature amplitude modulation (QAM), 8 multiple input multiple output (MIMO) layers, and 32 antenna ports, where the required bandwidths can be found in [6, Table 7-2]. These high bandwidth requirements have driven the standardization of new 25G and 50G PON systems. In preparation for the beyond 5G (B5G) and 6G in 2030, next-generation wireless and mobile services are expected to impose even higher bandwidth demands on the optical access network. Bandwidth requirements are expected to continue to rise to 100G and beyond. Therefore, mobile x-haul will be a driver for the next-generation 100G-and-beyond PONs.

Business Services and More: Business users are high-end users in the broadband access network that play a key role in determining the peak service tier and the peak network capacity. Busi-

ness users can be enterprise buildings, industry campuses, or high-bandwidth consumers that provide cloud services. The broadband industry usually uses Nielsen's Law of Internet Bandwidth to predict broadband Internet speeds for high-end users, where a 50 percent compound annual growth rate (CAGR) is typically used to forecast the growth in the downstream speed. Considering the current 10G PON (with a 5 Gb/s peak service tier in download speed, assuming a 2:1 capacity-to-peak tier ratio), 100G capacity will be required to support peak service tiers from high-end users in the optical access network in approximately 2030 if the 50 percent CAGR predicted by Nielsen continues [7]. Therefore, to keep pace with Nielsen's Law, a 100G PON will probably be needed by 2030.

ADVANTAGES OF COHERENT PONs

One of the most significant motivations for introducing coherent detection into 100G-and-beyond single-wavelength PONs is that the single-wavelength IM/DD PON will reach its sensitivity limit above 50 Gb/s speeds. The typical PR-30 power budget defined in the IEEE and ITU-T PON standards is 29 dB; however, in practice, operators usually require a loss budget of more than 31 dB to coexist with legacy infrastructures and reserve an additional loss budget for field operations. The ITU-T standards define 32 dB and even 35 dB as requirements for long-reach PON systems for extended power budgets.

IEEE 802.3ca NG-EPON defines a receiver sensitivity of 25G per wavelength as approximately -24.1 to -24.3 dBm at a bit error rate (BER) level of 1×10^{-2} . The full specifications of the ITU-T 50G PON are still in development, and early phase results show the sensitivity increases to -22.75 dBm at a BER level of 1×10^{-2} . Better results have generally been obtained in lab tests. For example, a -22 dBm sensitivity was demonstrated for a single-wavelength 100G PON based on an O-band IM/DD with semiconductor optical amplifier (SOA), APD, and more than 189 nonlinear equal-

ization taps. In addition to complex digital signal processing (DSP), a high launch power is needed for 100G PON to stay within the required power budget. Achieving such a high launch power at the optical network unit (ONU) side is still too challenging in the upstream. In addition, fiber dispersion causes significant penalties for high-baud-rate signals. Coexistence with traditional PON services makes it challenging to find a suitable transmission wavelength window in the O-band. Therefore, for increasing the data rate for a single wavelength, limited sensitivity will become too challenging for the 100G TDM-PON to meet the PR-30 power budget (> 31 dB link budget in practice) by using direct detection in the O-band, not to mention extended power budgets.

Coherent optics has been widely deployed in long-haul and metro transmission systems and has now been introduced into access networks due to its superior receiver sensitivity and the maturity of the technology. Coherent optical detection can be simply described as beating between an incoming signal and a clean local oscillator (LO) signal, as shown Fig. 3a. Employing coherent optics in a PON offers four advantages, as described below.

Superior Receiver Sensitivity: Coherent beating between the signal and a clean LO significantly improves the receiver sensitivity. Figure 3b compares the sensitivity for coherent detection and the IM/DD for 100G-and-beyond access networks. The receiver sensitivity for coherent detection of a 100 Gb/s polarization-division multiplexing quadrature phase shift keying (25-Gbaud PDM-QPSK) signal is more than 14 dB better at a BER of 1×10^{-3} than a 100 Gb/s IM/DD system that uses transceivers with a 100 Gbaud non-return-to-zero (NRZ) signal with 100 GHz bandwidth. Note that 100 GHz optics is still very challenging. If we limit the transceiver optical bandwidth to 25 GHz, the sensitivity improvement by coherent detection is more than 19 dB, compared to 100 Gbaud NRZ or 50 Gbaud pulse amplitude modulation (PAM)-4 signals. Simulation results show that the sensitivity of coherent optics is far better than the IM/DD solutions for bandwidth limited cases. For higher speeds, the simulated sensitivity of the 200G single-wavelength PON based on 50 Gbaud PDM QPSK can be as low as -36 dBm at a BER of 1×10^{-3} and -38.5 dBm at a BER of 1×10^{-2} . Therefore, a coherent PON can support a longer transmission distance and/or a larger number of end users than an IM/DD PON.

High Speed Realization by Using Advanced Modulation Formats: Multidimensional and more advanced modulation formats of coherent optics enable high-speed access [8]. Unlike a DD solution in which the signal is represented in only one dimension by the intensity or amplitude, coherent optics enables linear conversion of the optical field and can detect signals on the amplitude, phase, and polarization. Therefore, multidimensional modulation can be achieved to enable high spectral efficiency and consequently high-speed data transmission based on low-bandwidth optoelectronic devices. An example is shown in Fig. 3b: 100 Gb/s can be achieved using PDM-QPSK and 200 Gb/s can be achieved using PDM-16-ary quadrature amplitude modulation (16-QAM) for the same 25G class of optoelectronic devices.

Coherent optics has been widely deployed in long-haul and metro transmission systems and has now been introduced into access networks due to its superior receiver sensitivity and the maturity of the technology.

The simulated sensitivity of the 200G single-wavelength PON based on 25 Gbaud PDM 16-QAM can be as low as -32 dBm at a BER of 1×10^{-3} and -34.5 dBm at a BER of 1×10^{-2} , which is much better compared to IM/DD solutions.

Digital Dispersion Compensation and More:

Another significant benefit of coherent detection is that advanced DSP is enabled by linear conversion of the optical field. Optical and electrical impairments (e.g., chromatic dispersion) can be completely handled in the digital domain [8, 9]. This property of coherent detection is very attractive and useful because high-speed access in the C-band is enabled. Compared to alternative IM/DD PONs in which wavelengths are allocated to the O-band to prevent incurring a dispersion penalty, the wavelength window is considerably expanded using coherent detection because chromatic dispersion is almost eliminated. Consequently, more optical spectra become available for future use and network upgrades.

Channel Selection without an Optical Filter:

Finally, coherent detection provides the capability of frequency selection without the need for an optical filter. The inherent frequency selectivity of coherent detection enables the detection of only frequencies close to the LO frequency for a receiver with limited bandwidth; or if a broadband coherent receiver is used, digital channel selection can be realized by using digital filters. Therefore, the wavelength of the LO in the OLT or ONU can be changed to select a specific channel or subchannel. Consequently, no optical filter is needed at the receiver side, and no frequency-selective components are required in the optical distribution network (ODN). This characteristic is useful for future WDM coherent PON extensions. Furthermore, coherent detection also enables the additional frequency-division multiplex (FDM) scheme, which is discussed later.

The analysis presented above shows that to fulfill a PR-30-and-beyond power budget, a 100G single-wavelength PON will probably require coherent detection. Therefore, it has attracted lots of research attention recently as a promising candidate for 100 Gb/s or higher PON systems. There have already been many experimental demonstrations of coherent PONs achieving 100 Gb/s for single wavelengths over 50 and 100 km by using adaptive DSP for chromatic dispersion compensation, clock recovery, frequency offset compensation, polarization demultiplexing, carrier phase recovery, and channel equalization. An 800-G λ (T) WDM coherent PON based on 100G/ λ has been used to demonstrate the considerable potential of coherent detection for future terabit-per-second PON super broadband access [10].

CHALLENGES

Despite the numerous advantages offered using coherent optics in next-generation PON systems, major challenges associated with coherent PONs remain. In this section, we discuss issues that need to be resolved to promote the use of coherent

Upstream burst-mode detection is the second major challenge to the implementation of a coherent single-wavelength TDM PON. In the PON downstream, signals are continuously broadcast to all users, whereas the signals in the upstream are received from different users, burst by burst.

optics in access networks. A considerable amount of research is being carried out on these issues.

COST, COMPLEXITY, AND POWER CONSUMPTION

Cost is the first challenge that must be resolved to introduce coherent optics into access networks [8, 9]. A PON is generally directly connected to end users and is therefore highly sensitive to the cost of the optical and electrical components used in the network because of the enormous market involved. Coherent optics used in long-haul and metro optical transmission are all typically based on state-of-the-art photonic and electrical components to obtain best-in-class performance. Direct implementation of coherent optics from long hauls and metros in access networks is uneconomical because of high costs. In addition, compared to an IM/DD PON, coherent PON requires additional LO lasers, advanced modulators, additional photodetectors (PDs), digital-to-analog converters (DACs)/analog-to-digital converters (ADCs), and DSP application-specific integrated circuits (ASICs). These components increase the transceiver cost and complexity, as well as the power consumption. Therefore, the lasers, modulators, coherent receivers, and DSPs in an access network must be optimized to lower the complexity, cost, and power consumption.

UPSTREAM BURST-MODE DETECTION

Upstream burst-mode detection is the second major challenge to the implementation of a coherent single-wavelength TDM PON. In the PON downstream, signals are continuously broadcast to all users, whereas the signals in the upstream are received from different users, burst by burst. The upstream signals may have completely different power levels, clocks, frequencies, phases, and states of polarization (SOPs). Compared to the burst-mode receiver in IM/DD, coherent burst-mode detection is even more challenging due to the complexity of coherent optical signals. These challenges include burst-mode linear amplification and burst-mode DSP. As the upstream signals have different power levels, the first challenge of burst-mode coherent detection is burst signal amplification. For an IM/DD PON, burst-mode amplification is typically realized by using a burst-mode limiting transimpedance amplifier (BM-LM-TIA) after the PD. However, for coherent burst-mode detection, multiple linear and identical burst-mode TIAs are required for two or four paths in polarization- and phase-diversity coherent detection. Burst-mode DSP is another issue that needs to be solved. Conventional DSP algorithms used in coherent detection are typically designed for point-to-point (P2P) continuous mode receivers and are not suitable for burst-mode detection because of an excessively long acquisition time for signal recovery. Burst-mode DSP requires rapid action by all subsystems to detect short bursts. Additionally, the frequency-offset compensation range in the upstream burst-mode coherent

detection should be sufficiently large to cover possible wavelength drifts from different users.

ENABLING TECHNOLOGIES

SIMPLIFICATION AND OPTIMIZATION OF COHERENT OPTICS

To adapt coherent optics to access networks, many methods are available to simplify the setup, optimize the DSP, and reduce power consumption. Simplification and optimization of coherent optics in the access network have currently become active research topics in both industry and academia [11]. The cost of coherent detection can be broken down into several parts: the laser source, transmitter, receiver, and DSP. The first three parts are mainly optical and electrical components that can be simplified in many ways.

In long-haul transmission, laser sources on both the transmitter and receiver sides are generally expensive, high-quality, narrow-linewidth (typically less than 100 kHz) external cavity lasers (ECLs). Typically, two laser sources are needed in one transceiver, including one signal carrier and one LO. One solution for cost reduction is to use fewer lasers in the ONU. Using a remote laser source from the centralized OLT for each ONU would reduce the overall cost by distributing the laser cost over the whole network [4]. Another way of lowering the cost would be to replace expensive ECLs with lower-cost lasers, such as distributed feedback (DFB) lasers (~ 1 MHz linewidth), and allow an acceptable power penalty in the access network [8].

Another major simplification is related to the transceiver, especially on the receiver side. A full-field optical coherent receiver is typically used in long-haul transmission systems and consists of four pairs of balanced PDs (8 PDs) with polarization- and phase-diversity homodyne coherent detection. Consequently, four ADCs are required for signal processing. This setup could be simplified by using fewer PDs. The use of heterodyne coherent detection can halve the number of required balanced PDs, ADCs, and associated TIAs. The system could be further simplified by removing the polarization-diversity receiving system and using polarization scrambling or a special polarization-time block coding method [11]. In addition to simplifying the receiver, coherent detection could also be applied using conventional intensity or amplitude modulation formats, for example, NRZ or PAM-4. This modification would also simplify the transmitter side without the use of complex in-phase and quadrature (I/Q) modulators.

DSP optimization in a coherent PON is also a very active research area in which system functionalities are optimized to reduce power consumption [8]. Compared to long-haul and metro coherent systems in which distance-dependent equalizations result in considerable computational complexity and power consumption, the access network generally requires much less DSP due to the shorter transmission distance. For example, fixed chromatic dispersion with huge computation complexity can be eliminated in the access network, as the dispersion accumulating in a typical access link can be compensated in the adaptive equalizers with a small number of taps. The lower number of digital taps represents lower power consumption from the DSP. In addition, polar-

ization mode dispersion (PMD) compensation is not necessary over a 20 km fiber link, which further reduces both the complexity and power consumption. Other functions, such as carrier phase recovery, can be simplified by using shared phase estimation information between two polarizations for which the phase noise is almost identical in short transmission links [10]. These modifications can significantly reduce the total energy consumption by optimizing the DSP for the access network environment [8].

COHERENT BURST-MODE DETECTION

As mentioned above, a key challenge for coherent PON is the realization of upstream coherent burst-mode detection. Many innovations have been made in this research area.

Burst-mode coherent detection generally includes burst-mode linear amplification and burst-mode DSP. Burst-mode amplification is required to deal with different signal power levels and enlarge the dynamic range. As considerable challenges are still associated with coherent receivers with burst-mode linear TIAs, optical burst-mode preamplification is an effective method of leveling burst signals before these signals are sent to coherent receivers. In [12], a real-time burst-mode coherent receiver with an automatic gain control SOA (AGC-SOA) was demonstrated and a real-time digital signal processor was proposed. Most recently, a 25 Gbaud PAM-4 linear burst-mode receiver designed in a 0.25 μm SiGe:C was demonstrated, showing the possibility of using burst-mode linear TIAs in the future.

Burst-mode DSP and an efficient preamble design at the receiver side are needed in addition to optical amplification. The acquisition time for signal recovery at the coherent burst-mode receiver must be sufficiently short to improve the upstream efficiency. Specially designed preambles and fast DSPs have been reported. In pilot-aided time-domain estimation, a low-complexity widely linear compensation method is used to compensate for the IQ imbalance, and burst-mode DSP based on precalculated finite impulse response (FIR) filter coefficients has also been demonstrated to shorten the acquisition time.

Figure 4 is a schematic showing the principle of the burst frame structure and upstream burst-mode signal recovery functions for a TDM coherent PON [13] with different synchronization patterns. Although the general concept is taken from an E-PON, most of the preamble patterns and related subfunctions are specially designed. Here, we consider the most complex case in which the optical signals are modulated and multiplexed on the phase, polarization, and amplitude. The proposed preamble consists of three sync patterns, SP-1, SP-2, and SP-3. Based on nearly equally distributed QPSK symbols, SP-1 is designed for receiver settling with burst-mode amplifications. SP-2 is designed for burst clock recovery. SP-3 is designed for data-aided DSP synchronization functions, including frame synchronization, synchronization, and frequency synchronization (frequency offset estimation). In [13], $2 \times N$ conjugate symmetric symbols are used to demonstrate the feasibility of efficient preamble design and signal processing, where three DSP functions share the same preamble to reduce

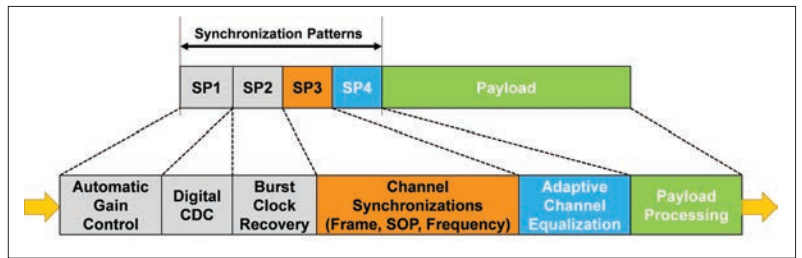


FIGURE 4. Coherent burst frame design with burst signal processing (SP: synchronization patterns; CDC: chromatic dispersion compensations; SOP: state of polarization).

the overall preamble length. Based on low-order training symbols (e.g., QPSK symbols), SP-4 is designed for adaptive channel equalization. Based on the information of the SOPs estimated by SP-2, the inverse of the Jones matrix can be utilized to reduce the convergence time of adaptive channel equalization. Finally, the payload process can be simplified using the information from the proposed preamble. High-order modulation formats can be used in the payload block. Finally, the order of these preamble patterns and burst-mode DSP functions may differ from that shown in Fig. 4, depending on the algorithms used.

FUTURE RESEARCH DIRECTIONS

Coherent optics provide many benefits to access networks beyond sensitivity improvement. Innovations will foreseeably continue to be made in this area that promote technology development and take full advantage of coherent detection. In addition to the above-mentioned key enabling technologies, there are some interesting research directions that have been attracting increasing interest.

FLEXIBLE MULTIPLEXING BEYOND TDM

The development of coherent detection will enable the utilization of more flexible multiplexing, including TDM, frequency-division multiplexing (FDM), WDM, and multidimensional multiplexing, in optical access networks. Solutions such as XR optics-enabled P2MP coherent optical network architecture utilizing digital subcarrier multiplexing technology show the possibility of FDM in a fine frequency granularity.

TDM has been the dominant multiplexing technology in optical access networks for decades, and WDM has been proposed to considerably increase the fiber capacity. Both coherent TDM-PON and coherent WDM-PON have been reported recently for next-generation access, showing the significant advantages of coherent detection. A TDM-PON can achieve a high peak data rate because of a high statistical multiplexing gain for bandwidth sharing with colorless components. A WDM-PON can provide secure virtual P2P and low latency to dedicated users. More flexible bandwidth allocation can be achieved for the next-generation access network by taking advantage of both types of PONs mentioned above, as shown in Fig. 5, where a novel rate-flexible dynamic TFDM 100G/ λ coherent PON architecture based on DSCM technology [14] is used to combine the advantages of TDM and FDM. Multiple receivers and different wavelengths are generally necessary for different subcarriers from different ONUs to prevent optical beating inter-

ference in IM/DD systems. However, whether downstream or upstream, the OLT and ONU require only one wavelength and one transceiver in the proposed TFDM system due to coherent multiple subcarrier detection. The TFDM coherent PON enables multiplexing flexibility in both the time and frequency domains for P2MP coherent access networks.

In addition, combining TDM, FDM, WDM, and polarization-division multiplexing (PDM) into one coherent system could enable multidimensional coherent access. Novel multiplexing schemes, including time-division multiple access (TDMA), frequency-division multiple access (FDMA), code-division multiple access (CDMA), and non-orthogonal multiple access (NOMA), and so on, are also worthy of further study in future coherent PON systems.

RATE-ADAPTIVE ACCESS

Currently, PONs deliver the same peak data rate to different ONUs. In such a system, the maximum data rate is limited by the ONU with the worst performance. Many ONUs have good channel

conditions, and there is a large gap between the transmitted data rate and the maximum channel capacity. A rate-adaptive system can easily solve this problem by flexibly adjusting the data rate of each ONU according to the channel conditions. To date, some rate-adaptive access technologies have been demonstrated in IM/DD PON systems based on hybrid modulation formats, constellation shaping, and variable FEC coding rates [15]. Compared to a conventional IMDD system in which only one dimension is used for signal modulation, coherent PON utilizes more dimensions, such as the polarization, amplitude, and phase. Therefore, more solutions would be available to exploit the adaptive data rate. For example, the adaptive LDPC-FEC code could be used to adjust the data rate.

Constellation design is an alternative method to explore in rate-adaptive coherent access. For example, probabilistic shaping (PS) is expected to be a promising technology in coherent PON due to the granularity of the information rate and wide flexibility. PS can approach the optimal channel capacity and take full advantage of the high transmission speed of the coherent system. We used PS to simulate the performance of rate-adaptive access in a 20 km coherent network, and the results are shown in Fig. 6. The net data rate changes continuously as the received power decreases. The entropy of the transmitted signal depends on the channel condition, namely, the effective signal-to-noise ratio (SNR). PS can theoretically cover an ultra-large range of received power until the QPSK signal fails. During this process, the modulation formats change from 64-QAM to QPSK, whereas the net data rate decreases from 255 Gb/s to 85 Gb/s.

CONCLUSIONS

We have reviewed the use cases, challenges, and recent progress for using a high-speed coherent PON for 100G λ -and -beyond fiber access networks. An analysis of the bandwidth requirements for three major services shows how 100G λ -and-beyond PONs will be needed. In addition, four major advantages of using coherent optics to obtain 100 Gb/s-and-beyond single-wavelength fiber access are presented. As 100 Gb/s per wavelength and beyond will be required for next-generation access

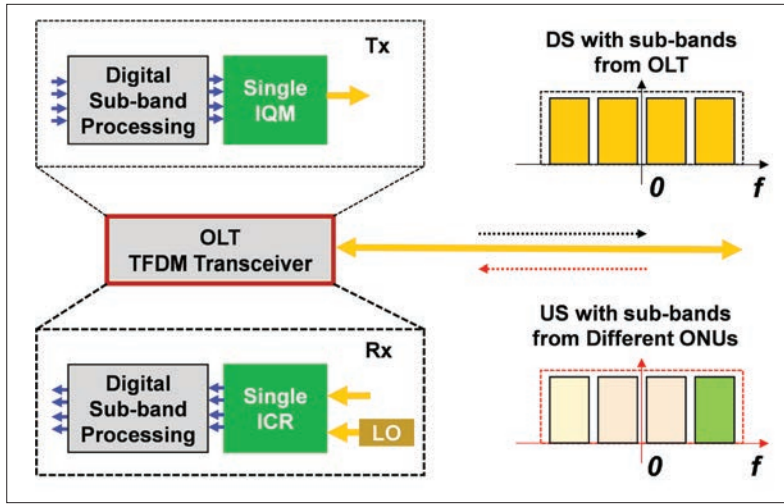


FIGURE 5. The principle of a TFDM coherent PON in which a digital sub-band is used for time and frequency two-dimensional resource sharing (Tx: transmitter; IQM: I/Q modulator; Rx: receiver; ICR: integrated coherent receiver; LO: local oscillator).

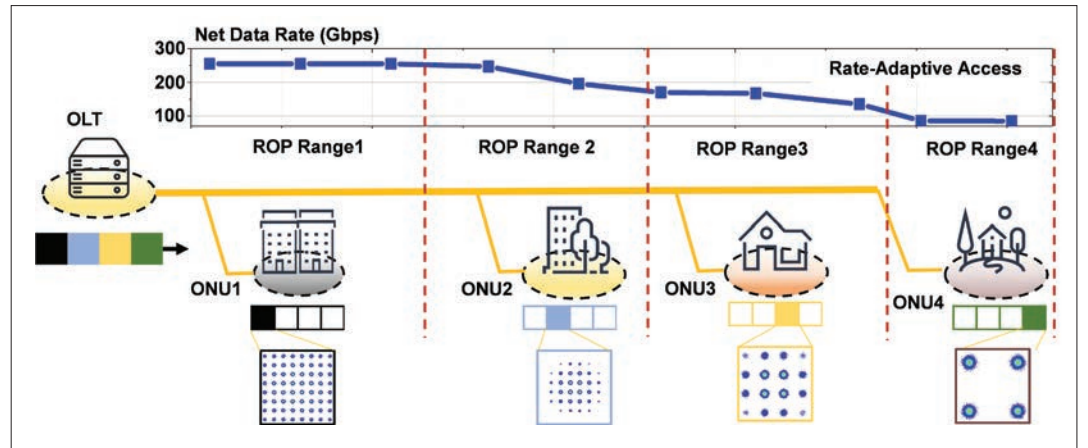


FIGURE 6. The principle of rate-adaptive access based on constellation shaping (ROP: received optical power); here, we assume that ROP Range 1 > ROP Range 2 > ROP Range 3 > ROP Range 4; the simulation setup is the same as in Fig. 3b, where 25 GHz coherent optics with full-field, polarization /phase diversity, and homodyne coherent detection is used.

networks to satisfy the continuously growing bandwidth demand, coherent optics is a promising candidate for realizing these rates.

However, major challenges to the implementation of coherent PONs remain. Cost, complexity, and high power consumption are among the challenges that must be resolved to introduce coherent optics into access networks. Upstream burst detection is another problem that must be solved to implement coherent TDM-PONs. Many schemes have been developed to adapt coherent optics to access networks. We review recent progress and present technical insights into key enabling technologies for coherent PONs, including coherent upstream burst detection, burst-mode DSP, and innovations for reducing costs, complexity, and energy consumption.

Note that the use of coherent optics provides benefits to access networks beyond sensitivity improvement. To take full advantage of coherent detection, future directions, including flexible multiplexing beyond TDM and rate-adaptive access in coherent PON, are introduced and discussed. It is foreseeable that further innovations will be made in this area to make coherent PONs a reality.

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BIOGRAPHIES

JUNWEN ZHANG [M'13] received a Ph.D. in the field of optical communication from Fudan University, Shanghai, China, in 2014. He is currently a full professor at Fudan University. He was at CableLabs from 2018 to 2020, focusing on optical and wireless access technologies. He has published more than 200 papers on the topics of high-speed optical and wireless communication systems. He has received a Marconi Society Paul Baran Young Scholar Award, an IEEE Photonics Society Graduate Student Fellowship, and a Wang Daheng Optics Award. He is currently the Subcommittee Chair of the OFC2022 Technical Committee at Optical Access Networks.

ZHENSHENG JIA [S'19] is a Distinguished Optical Technologist at CableLabs. He has published more than 180 peer-reviewed journal and conference papers, one book, and 10 book chapters. He also holds more than 100 U.S. and international granted patents. He is a Fellow of Optica (formerly Optical Society of America). He holds Bachelor's and Master's of Science degrees in electronic engineering from Tsinghua University, Beijing, China, and a Ph.D. in electrical and computer engineering from Georgia Institute of Technology, Atlanta.